Iris Liu

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Education

Carnegie Mellon University — Pittsburgh, PA
Master of Educational Technology and Applied Learning Science, School of Computer Science — *Expected Dec. 2026*
B.S. Cognitive Science and Human-Computer Interaction — *May 2026*

- Selected Coursework: Experimental Design for Behavioral and Social Sciences, Research Methods: Meta-Analysis, User Centered Research and Evaluation, ML in Production, Database Management

Technical Skills

Research Methods: Experiment Design, Causal Inference, Quasi-Experimental Design, Inter-Rater Reliability, Mixed Methods, Qualitative Coding, User Studies, Interrupted Time Series, Meta-analysis, A/B Testing, Survey Research, UX Research
Languages: SQL, Python, R, JavaScript, Java, C
Data: Pandas, NumPy, Statistical Modeling, Data Visualization
ML/AI: LLM Evaluation, Model Evaluation, Recommender Systems, Prompt Engineering, RAG, NLP, LLMs, PyTorch, Scikit-learn
Tools: Git, Linux, AWS, Kafka, SQLite, Figma, Ollama, HTML/CSS, Claude Code, GitHub Copilot

Experience

Learning Analytics Consultant — Jan. 2026 – Aug. 2026
Learnvia — *Pittsburgh, PA*

- Led a 5-person team through a learning analytics study design including scoping requirements, designing a 13-category behavioral taxonomy, and validating results to deliver a methodology for analyzing 58,000+ student-AI interaction records
- Directed development of a privacy-preserving LLM-assisted coding workflow, defining requirements and coordinating implementation in Python with local models (Ollama) and AWS
- Designed a human-review validation procedure comparing LLM-generated labels against human raters (κ up to 0.87), delivering a reusable evaluation tool and methodology recommendations to the client

Statistics Intern — May 2026 – Present
CATS Lab, Carnegie Mellon University — *Pittsburgh, PA*

- Built reproducible R pipelines to clean, integrate, and verify log data from randomized classroom studies spanning 30+ classrooms, resolving implementation-fidelity issues where recorded conditions diverged from actual tool use
- Developed rules to distinguish genuine intervention exposure from non-use and teacher testing, defining the analytic sample and documenting decisions for reproducibility across the research team
- Applied interrupted time series and longitudinal methods to estimate intervention effects, and produced exploratory visualizations of student engagement and learning outcomes across conditions

Research Assistant — June 2025 – Aug. 2025
McLearn Lab, Carnegie Mellon University — *Pittsburgh, PA*

- Built learning curves from student performance data in educational math games and refined knowledge component mappings, determining which skills each activity actually measured
- Presented poster on computational models of student learning and practice scheduling at CMU LearnLab Summer School

UX Design, Development and Analytics Intern — June 2024 – Aug. 2024
Biomotivate — *Pittsburgh, PA*

- Built JavaScript tooling integrating LLM APIs to streamline narration generation and video editing workflows
- Ran VR user studies with 10+ participants and analyzed wearable sensor data to inform development of personalized emotional resilience tools
- Designed and implemented mobile/desktop interfaces through iterative user research and prototyping in Figma/HTML/CSS

Research and Projects

LLM Labeling + Evaluation Pipeline for Human-AI Tutoring Dialogue | GitHub — Jan. 2026 – Aug. 2026

- Open-source Python pipeline applying 13 validated behavioral codes to 58k turns of student–tutor dialogue using a local LLM (Qwen3-8B via Ollama), keeping FERPA-protected transcripts on-premise for privacy compliance
- Validated every code against a human gold standard using precision, recall, F1, and Cohen's κ, raising student-code agreement from κ 0.20–0.66 to 0.67–0.87 (tutor codes 0.66–1.00) through case-level analysis of LLM–human disagreements
- Diagnosed a silent failure where hidden model reasoning exhausted the token budget and empty outputs defaulted to negative labels, eliminating parse failures and cutting per-message latency 10x
- Engineered a fault-tolerant pipeline with async batching and SQLite checkpointing for resumable multi-day runs, combining the LLM with rule-based heuristics (similarity matching, phrase rules, screening filters; κ = 1.0 on rule-based checks)

ESME: AI Teaching Assistant for Programming for Lovers — Jan. 2026 – May 2026

- Led research for a 5-person team prototyping a retrieval-grounded AI TA for a CMU online CS course serving 8,200+ learners
- Ran contextual inquiry and think-aloud tests with 11 students, plus a competitive analysis of 7 AI learning assistants
- Translated key insights on student distrust of chatbots into hint-first behavior, and contributed to the RAG pipeline to ground answers in course content